

Project Design Phase-II

Data Flow Diagram & User Stories

Date	17 July 2026
Team ID	
Project Name	SB Stocks – Stock Trading Application (MERN Stack)
Maximum Marks	4 Marks

Data Flow Diagram:

The diagram below shows how data flows through the SB Stocks trading platform — from the trader in the browser, through the Node.js/Express processes (Authentication, Market Data Engine, Trade Execution, Portfolio & Wallet Management, and History & Audit Log), into the MongoDB data stores.

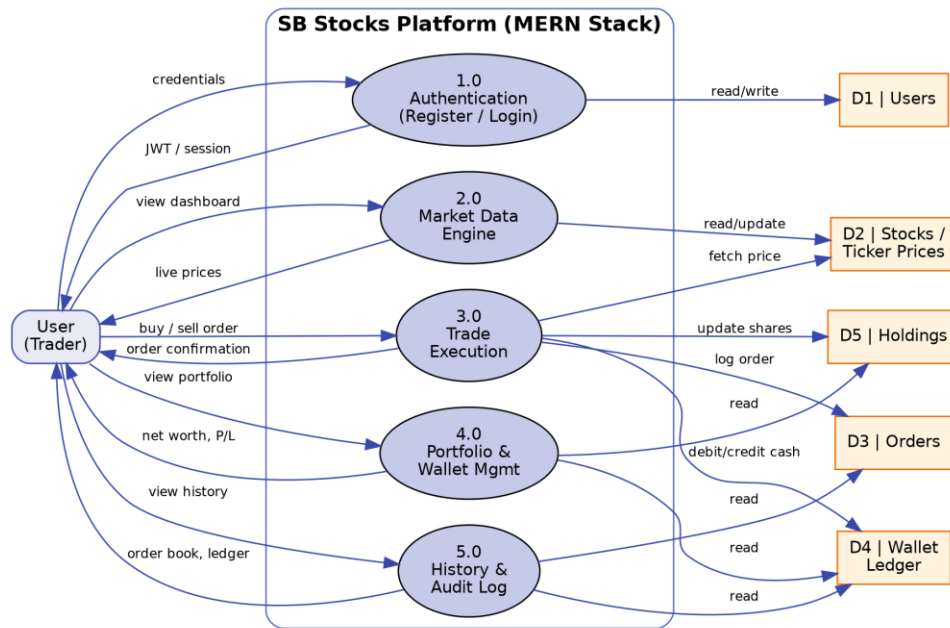


Figure 1: Data Flow Diagram (Level 1) – SB Stocks Trading Platform

User Stories

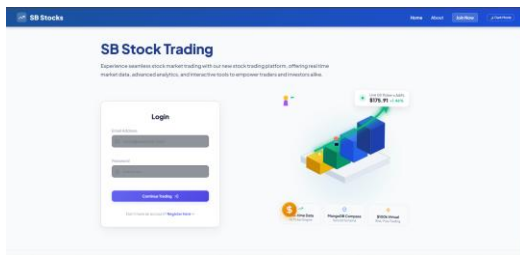
The table below lists the user stories captured for the SB Stocks trading platform.

User Type	Epic	Story #	User Story / Task	Acceptance Criteria	Priority	Release
Trader (End User)	Registration	USN-1	As a user, I can register by entering my name, email, and password.	I can access my dashboard after registering.	High	Sprint-1
	Registration	USN-2	As a user, I receive a confirmation email once I have registered.	I receive a confirmation email.	Medium	Sprint-1
	Login	USN-3	As a registered user, I can log in using my email and password.	I am redirected to my dashboard on success.	High	Sprint-1
	Market Dashboard	USN-4	As a trader, I can view real-time prices of stocks on my dashboard.	Prices auto-refresh every 4 seconds.	High	Sprint-1
	Market Dashboard	USN-5	As a trader, I can see Top Gainers and Top Losers at a glance.	Widgets show the correct top stocks by % change.	Medium	Sprint-2
	Stock Trading	USN-6	As a trader, I can view a stock's price chart before trading.	Chart renders Candlestick/Trend views for 1D/1W/1M/1Y.	High	Sprint-2
	Stock Trading	USN-7	As a trader, I can buy shares of a stock at the live market price.	Order executes; holdings and wallet balance update.	High	Sprint-2
	Stock Trading	USN-8	As a trader, I can sell shares that I own.	Order executes; cash is credited to my wallet.	High	Sprint-2

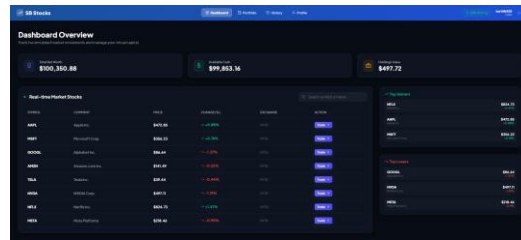
User Type	Epic	Story #	User Story / Task	Acceptance Criteria	Priority	Release
	Portfolio	USN-9	As a trader, I can view my holdings and overall profit/loss.	Portfolio page shows accurate valuation and P/L.	High	Sprint-3
	Wallet	USN-10	As a trader, I can deposit virtual funds into my wallet.	Wallet balance increases immediately after deposit.	Medium	Sprint-1
	Wallet	USN-11	As a trader, I can withdraw funds from my wallet.	Wallet balance decreases immediately after withdrawal.	Low	Sprint-3
	History	USN-12	As a trader, I can view my complete order book and wallet ledger.	History page lists all trades and transactions with timestamps.	Medium	Sprint-3
	Profile	USN-13	As a trader, I can view my account profile and role.	Profile page shows username, email, and role.	Low	Sprint-1
Administrator	Monitoring	USN-14	As an administrator, I can monitor overall user trading activity.	Admin can view aggregated trade/user data.	Low	Sprint-4 (Planned)

Application Screenshots

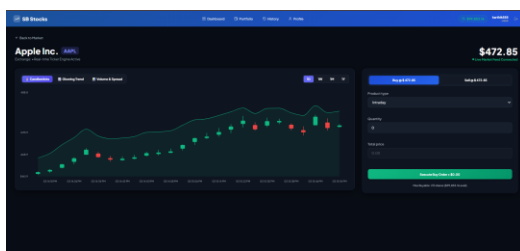
The screenshots below of the working application map to the user stories above, two per page for easy side-by-side review.



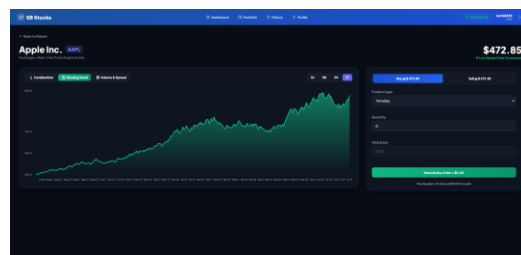
Login / Registration — USN-1, USN-2, USN-3



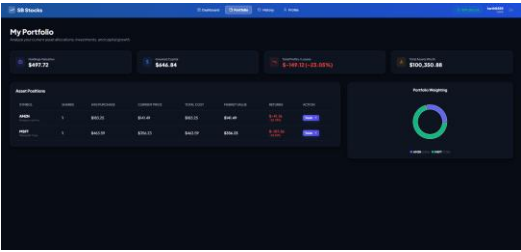
Dashboard Overview — USN-4, USN-5



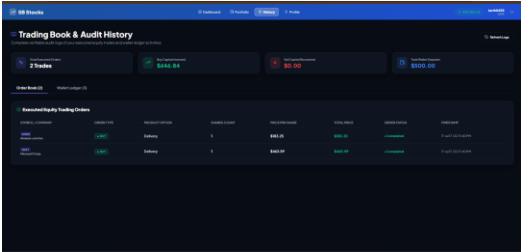
Stock Detail (1D Candlestick + Buy/Sell) — USN-6, USN-7, USN-8



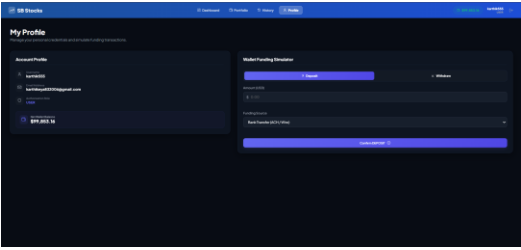
Stock Detail (1Y Trend) — USN-6



My Portfolio — USN-9



Trading Book & Audit History — USN-10, USN-11, USN-12



My Profile — USN-13